



StepperTune

PROFESSIONAL

USER MANUAL

From melodies to musical G-code
for Bambu Lab printers

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1. Introduction

StepperTune Professional converts manually written melodies or MIDI tracks into M1006 G-code commands, which supported Bambu Lab printers use to play jingles through their stepper motors. The application provides an audio preview on the computer and automatically applies the correct sound driver for the selected printer model.

IMPORTANT

StepperTune generates only the musical block to be inserted into the printer preset. It does not replace the normal printing G-code and must not be used to overwrite the entire Machine start G-code or Machine end G-code section.

Main features

- Manual composition using Note@Duration syntax.
- Multi-track MIDI import.
- Monophonic mode for all supported models.
- Polyphonic mode with up to three voices on compatible CoreXY printers.
- Individual or simultaneous preview, per-voice volume and Auto Balance.
- Start and End jingle generation and .gcode export.
- Complete project saving in the .stf format.

2. Interface and operating modes

Monophonic mode keeps the entire workflow in a single workspace. The following screen shows the complete control layout with an A1 printer selected.

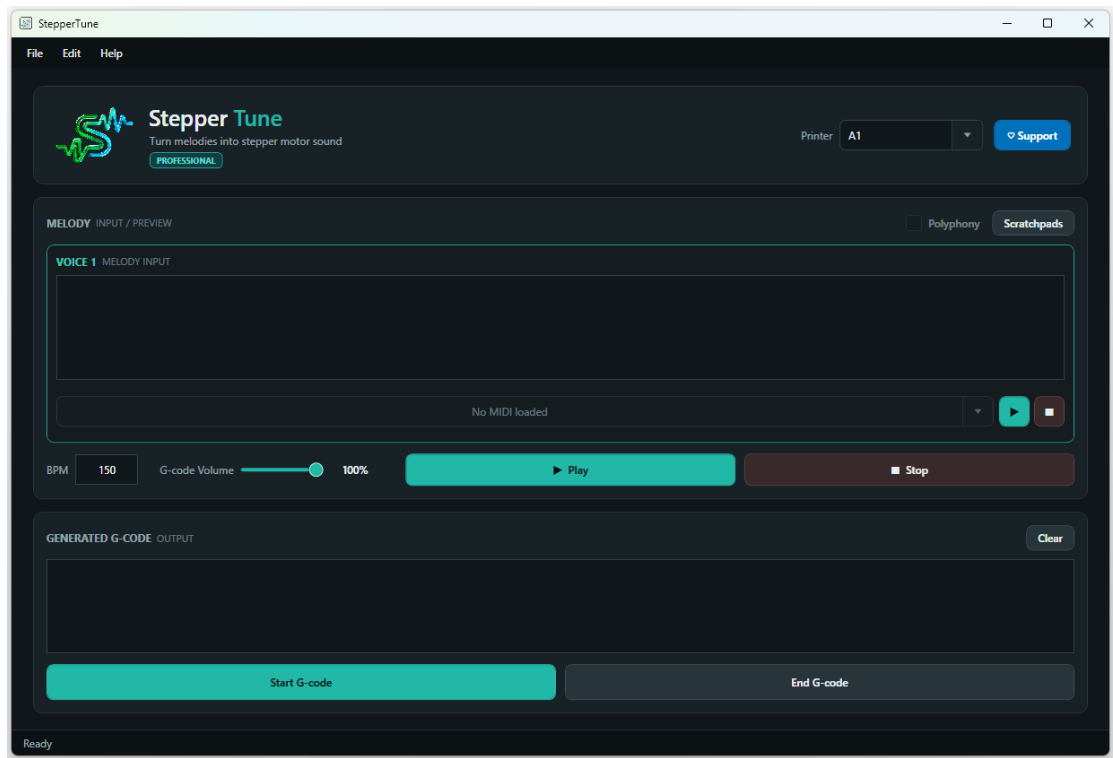


Figure 1 - Interface in monophonic mode with the A1 printer selected.

2.1 Control map

Area	Function
Menu File	New project, MIDI import, .stf loading/saving and G-code export.
Menu Edit	Loads the default Start or End sound G-code for the selected model.
Menu Help	Manual, update check and application information.
Printer	Selects the printer model; its driver determines G-code structure and parameters.
Melody Input	Contains the melody used to generate G-code.
MIDI Track	Assigns a MIDI track to the current voice.
BPM	Sets the melody tempo.
Generated G-code	Displays the generated sound block; Clear erases its contents.

2.2 Polyphonic mode

On compatible CoreXY models, Polyphony turns the editor into a three-voice workspace. Each panel has its own MIDI track, preview and volume level.

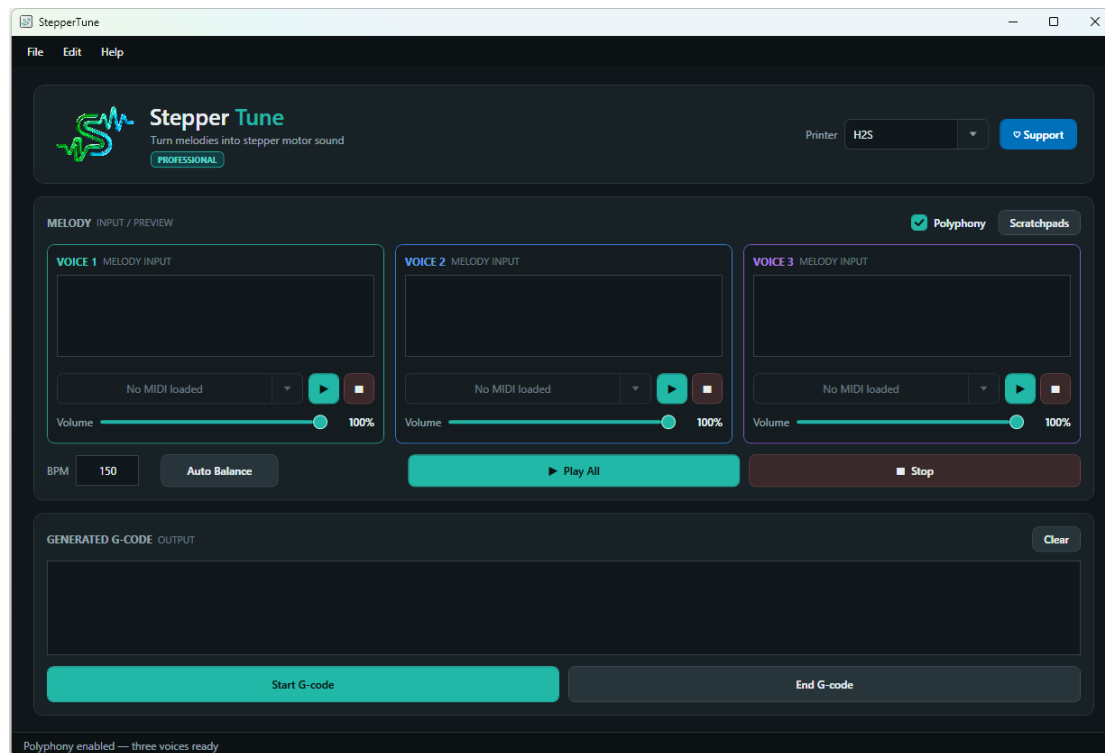


Figure 2 - Polyphony enabled: three independent voices, each with track selection, Play/Stop and volume.

2.3 How polyphony works

The Polyphony option is available only on compatible CoreXY models. When enabled, the editor is divided into Voice 1, Voice 2 and Voice 3. Each voice can contain a different melody or receive an independent MIDI track.

2.4 Playback controls

- The ► button beside a track plays only the corresponding voice.
- Play previews the active melody in monophonic mode; Play All plays all three voices together.
- Stop ends the combined playback; the ■ button for a voice stops its individual preview.
- During playback, the highlight follows the events being played.

NOTE

The preview uses the computer's audio system. The actual timbre depends on motor vibration and the printer's mechanical response, so it will sound different.

3. Writing and previewing a melody

Each event uses the Note@Duration format. Events are separated by a space.

EXAMPLE

```
C4@0.25 D4@0.25 E4@0.5 REST@0.25 G4@0.5
```

3.1 Notes, octaves and rests

- Available notes: C, C#, D, D#, E, F, F#, G, G#, A, A#, B.
- The number indicates the octave, for example C4 or F#5.
- REST inserts a pause and uses the same duration syntax, for example REST@0.25.
- Note names are not case-sensitive.

3.2 Duration and BPM

Values such as 1, 0.5, 0.25 and 0.125 represent relative musical durations. BPM determines the actual speed of the entire sequence. When a MIDI file is imported, StepperTune detects and proposes its BPM; the value can then be edited manually.

3.3 Selection and preview

- With no selection, Play starts from the beginning of the active editor.
- Double-click a note: Play starts from that note and continues forward.
- Select a continuous range of events: Play previews only the selection.
- The view scrolls automatically to keep upcoming events visible.

3.4 Scratchpad

The Scratchpads button temporarily replaces each Melody Input with its corresponding Scratchpad. You can test, cut, copy and rearrange fragments without changing the melody used for G-code generation.

KEY RULE

Scratchpad content can be previewed, but it is never used to generate G-code. Melody Input always remains the generation source.

4. Importing and assigning MIDI tracks

1. **Open the file** - choose File > Open MIDI and select a .mid or .midi file.
2. **Check the BPM** - verify the detected tempo and adjust it if necessary.
3. **Choose the track** - use the MIDI Track menu for the relevant voice. The track name and note count help identify its content.
4. **Assign the other voices** - in polyphonic mode, select a separate track for Voice 2 and Voice 3. A voice can also be left without a track.
5. **Preview** - listen to each voice separately first, then use Play All to evaluate the complete arrangement.

Choosing tracks

In complex MIDI files, not every track produces an effective result on the motors. Clear melodic lines with well-separated notes and registers that are not excessively low usually work best. Drum tracks, very dense chords or extensive overlaps can sound confused.

Recommended polyphonic workflow

- Voice 1: main melody or most recognizable part.
- Voice 2: accompaniment or counter-melody.
- Voice 3: bass, rhythmic accent or third line, usually at a lower volume.
- Remove or simplify parts that do not improve the acoustic result.

TIP

Do not try to reproduce every track from the original MIDI. Three carefully selected parts are usually clearer than an overcrowded arrangement.

5. Polyphony, volumes and Auto Balance

This example shows a complete project with three assigned MIDI tracks, different volume levels and an already generated Start G-code block.

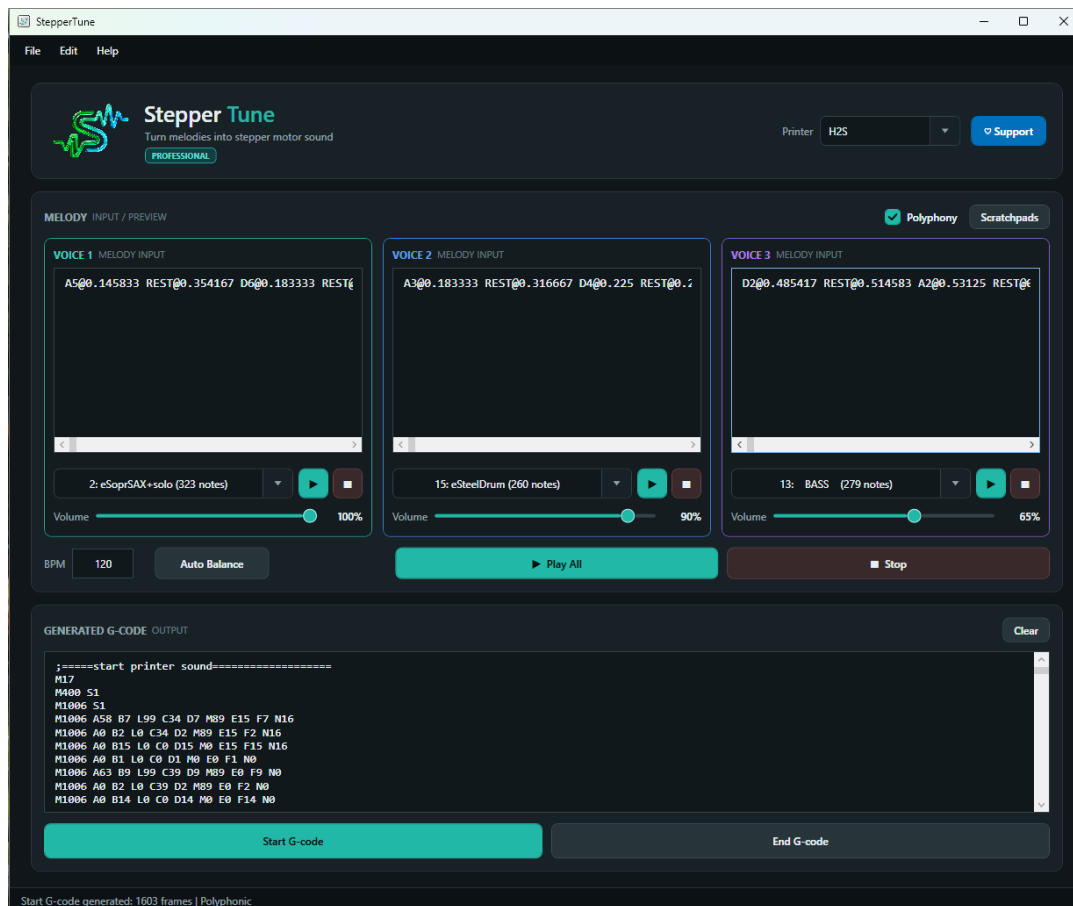


Figure 3 - Complete example: three assigned tracks, balanced volumes and generated Start G-code.

5.1 Monophonic volume

When Polyphony is disabled, G-code Volume controls the overall level of the generated melody.

5.2 Per-voice volume

In polyphonic mode, each voice has its own Volume slider. The value controls that part's contribution, allowing the main melody to stand out or the bass and accompaniment to be reduced.

5.3 Auto Balance

Auto Balance analyzes the musical presence of all three voices and calculates a conservative starting point for their levels. The result does not replace listening: resonances, note register and printer model all affect perceived volume.

PRACTICAL METHOD

Apply Auto Balance, listen with Play All and fine-tune the sliders by ear. If one voice masks the melody, reduce it gradually; if it disappears, raise it without automatically setting every voice to 100%.

6. Managing StepperTune projects

The native .stf format stores the state required to resume your work: printer, BPM, polyphonic mode, each voice's Melody and Scratchpad, assignments and the levels available in the current build.

Command	Result
File > New	Creates an empty project, closes the Scratchpads and clears MIDI data, selections and output.
File > Save Melody	Saves the current project to an .stf file.
File > Load Melody	Opens an .stf project and restores the saved work.
File > Export G-code	Saves the Generated G-code contents to a .gcode file.
Edit > Set default Start sound G-code	Loads the selected model's default Start sound into the output.
Edit > Set default End sound G-code	Loads the selected model's default End sound into the output.

Best practices

- Save the project regularly while preparing an arrangement.
- After changing the printer, check polyphony, levels and the generated G-code again.
- Keep the .stf project and the exported .gcode file separately.
- Use descriptive names that include the melody and target printer model.

7. Generating and exporting G-code

1. **Select Printer** - choose the exact printer model on which the jingle will be used.
2. **Prepare the voices** - write the melody or assign MIDI tracks. Enable Polyphony only when it is available and required.
3. **Check BPM and levels** - listen to each part and then to the complete arrangement.
4. **Generate** - press Start G-code for the startup sound or End G-code for the print completion sound.
5. **Check the output** - the block appears in Generated G-code; the status bar confirms the operation and mode used.
6. **Export** - choose File > Export G-code or copy the content while preserving every line break.

WARNING

Generate the G-code again whenever you change printer, BPM, melody, polyphonic mode or levels. Existing output is not updated automatically.

Start and End

The two buttons generate blocks intended for different preset sections. Do not swap them or paste both in the same location. Clear erases only the output area and does not change the melodies.

8. Adding the jingle in Bambu Studio

The following procedure is adapted from Bambu Lab documentation on MIDI/G-code sounds.

8.1 Open the printer preset

On the Prepare tab, select the correct printer and nozzle diameter, then click Edit preset beside the printer name.

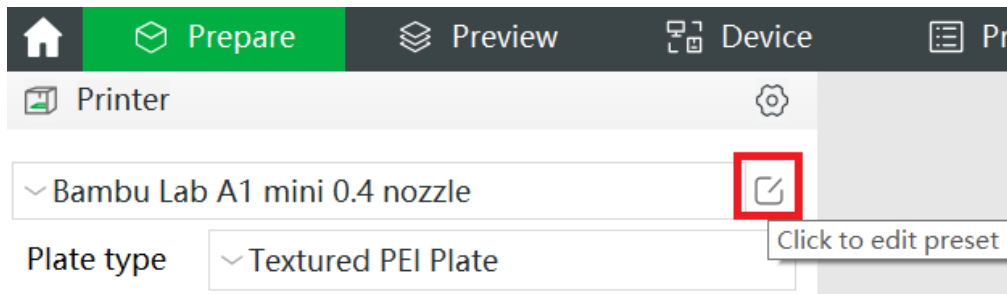


Figure 4 - Opening the printer preset. Source: Bambu Lab Wiki.

8.2 Open Machine G-code

The Machine G-code screen contains both the startup and print completion blocks. Edit only the indicated sound sections.

Enable Advanced

In the Printer settings window, enable Advanced at the top right, then open the Machine G-code tab.

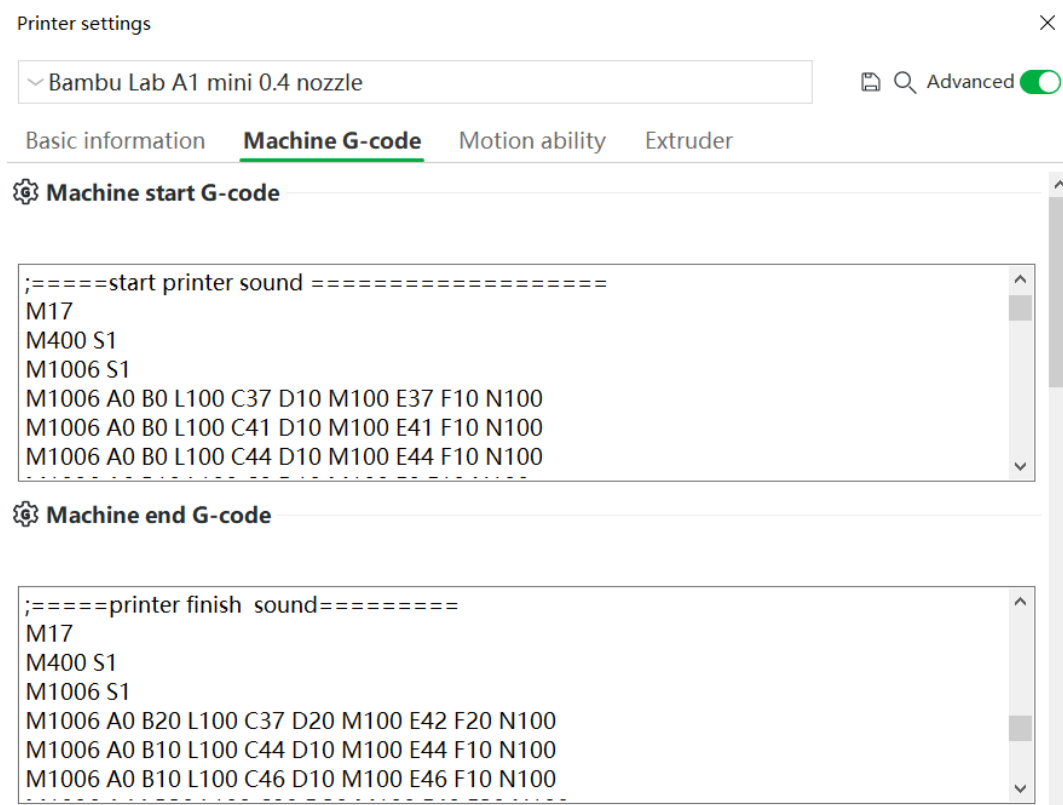


Figure 5 - Machine start G-code and Machine end G-code sections. Source: Bambu Lab Wiki.

8.3 Replace only the sound block

For the startup sound, locate the start printer sound separator in the Machine start G-code section. Replace only the code inside its sound block with the code generated by Start G-code.

For the print completion sound, locate the printer finish sound separator in the Machine end G-code section. Replace only the code inside its sound block with the code generated by End G-code.

DO NOT REPLACE THE ENTIRE SECTION

Leave every other command in Machine start G-code and Machine end G-code unchanged. Preserve the generated code's line breaks exactly.

8.4 Save the preset

Click Save. You can save the configuration as a User Preset, making it available to projects that use that preset, or as a Project Inside Preset, limiting it to the current project.

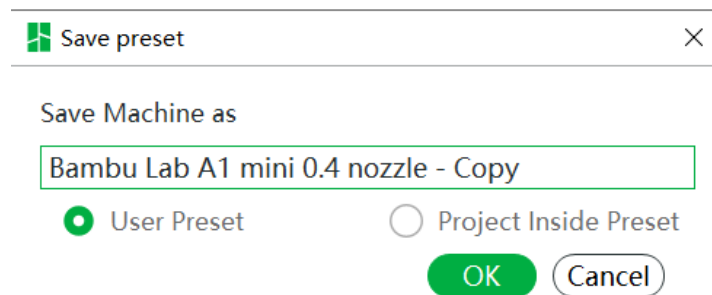


Figure 6 - Saving the preset. Source: Bambu Lab Wiki.

First test

- Keep a copy of the original preset.
- Perform the first test under supervision.
- Make sure the printer selected in StepperTune matches the printer used by the preset.
- After firmware or Bambu Studio updates, repeat a controlled test.

9. Supported printers and testing

Family	Models	Mode
Cartesian	A1, A1 mini	Monophonic
CoreXY	A2L, H2S, H2D, X2D, P2S	Monophonic and polyphonic with up to three voices

Always select the actual printer model being used. StepperTune automatically applies the G-code structure and acoustic parameters defined by its driver.

9.1 Physical test status

- Polyphonic playback verified on H2S and X2D.
- Monophonic playback verified on A1 and A1 mini.
- The other models are supported by their respective drivers but should be tested carefully, especially after firmware updates.

9.2 Pre-use checklist

- The selected printer model is correct.
- The preview contains no obvious errors or unwanted pauses.
- Voice levels are balanced.
- The G-code was regenerated after the latest change.
- The block was pasted into the correct section without deleting other commands.
- The original preset has been retained.

10. Safety, responsibility and references

StepperTune is an independent project and is not an official Bambu Lab product. It is not affiliated with, sponsored by or approved by Bambu Lab.

The software generates G-code commands that operate the printer motors to produce sounds. The user is responsible for selecting the correct printer model, checking the generated content and inserting it correctly into the preset.

Using manually modified G-code, running it on unsupported models or replacing sections other than the sound block is entirely the user's responsibility. Firmware, Bambu Studio and command behavior may change over time.

RESPONSIBLE USE

Stop the test if you notice abnormal movements, noises or vibrations. Restore the original preset if in doubt. The author cannot be held responsible for damage, failed prints, data loss or other effects resulting from improper use or manual modifications.

References

- Bambu Studio procedure and reference images: Bambu Lab Wiki - MIDI.
- Bambu Lab, Bambu Studio and the cited model names belong to their respective owners.
- StepperTune Professional - DGVeLab.

Support and updates

From the Help menu you can open the manual, check for updates and view information about the application. The Support button opens the page dedicated to supporting the project.